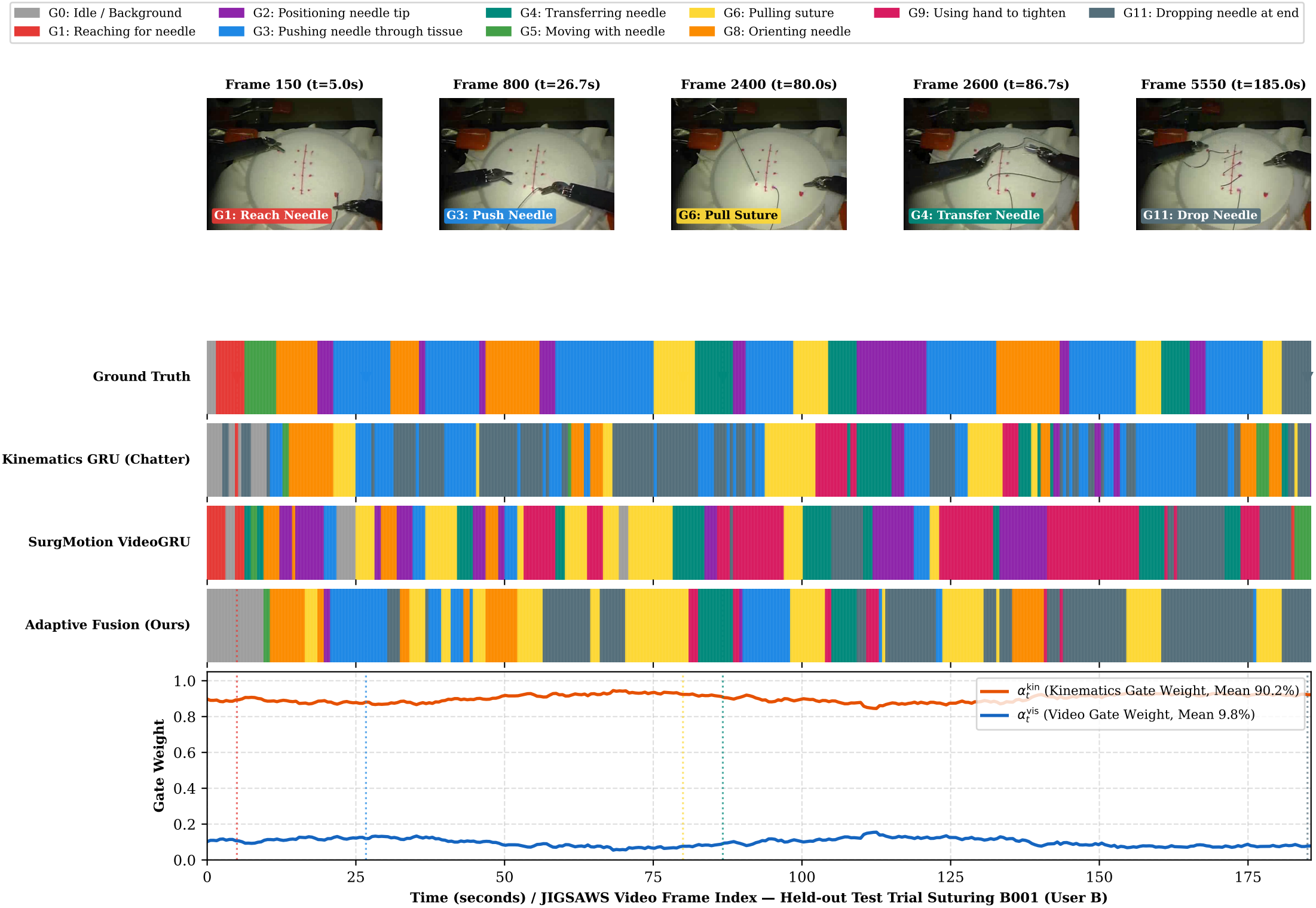

G0: Idle / Background
G1: Reaching for needle
G2: Positioning needle tip
G3: Pushing needle through tissue
G4: Transferring needle
G5: Moving with needle
G6: Pulling suture
G8: Orienting needle
G9: Using hand to tighten
G11: Dropping needle at end
Frame 150 (t=5.0s)
G1: Reach Needle
Frame 800 (t=26.7s)
G3: Push Needle
Frame 2400 (t=80.0s)
G6: Pull Suture
Frame 2600 (t=86.7s)
G4: Transfer Needle
Frame 5550 (t=185.0s)
G11: Drop Needle
Ground Truth
Kinematics GRU (Chatter)
SurgMotion VideoGRU
Adaptive Fusion (Ours)
Gate Weight
1.0
0.8
0.6
0.4
0.2
0.0
α_t^{kin} (Kinematics Gate Weight, Mean 90.2%)
α_t^{vis} (Video Gate Weight, Mean 9.8%)
0
25
50
75
100
125
150
175
Time (seconds) / JIGSAWS Video Frame Index — Held-out Test Trial Suturing B001 (User B)